

The Seven Roles of Artificial Intelligence: Potential and Pitfalls in Combating Misinformation

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Abstract

Misinformation can shape beliefs and undermine democracy, making effective mitigation essential. Generative AI presents risks and opportunities in this space—it can generate dubious content but also detect and counter misinformation at scale. AI’s ability to support and persuade people, facilitate discourse, and enhance media literacy underscores its potential. However, risks such as hallucinations, bias reinforcement, and manipulation highlight the need for responsible implementation.

This essay explores AI’s role as an Informer, Guardian, Persuader, Integrator, Collaborator, Teacher, and Playmaker, examining strengths, weaknesses, opportunities, and threats via SWOT analysis. For policymakers and technology leaders, we highlight the importance of regulations, transparency, human oversight, and AI literacy to ensure that generative AI can serve as a tool for truth rather than deception.

Keywords: artificial intelligence, SWOT analysis, disinformation, recommendations, risks, opportunities.

Introduction

Some people have expressed substantial concerns about the impact of AI on the health of the information environment, whereas others have focused on the potential benefits of AI-driven solutions to counter misinformation. This essay summarizes the available evidence, adopting a cautious analytical approach, given that research on AI systems and misinformation is still in its infancy. Using an updated SWOT framework (1), we categorise well-supported findings as *strengths* or *weaknesses*, and more speculative or emerging claims as *opportunities* or *threats*. Inspired by people’s tendency to anthropomorphise AI (2), we specifically discuss seven “roles” that AI may play in misinformation management.

Roles of AI in Misinformation Management

There are at least seven roles that AI could play in a complex information environment (see Table 1, below). AI can serve as an “Informer”, delivering rapid access to information; a “Guardian” that detects and flags misinformation; a “Persuader” that can help change beliefs and reduce misconceptions; an “Integrator” to facilitate debate and help people reach agreement; a “Collaborator” that can assist in research and critical thinking; a “Teacher” that offers individual feedback at scale; or a “Playmaker” to support gamified educational experiences that strengthen misinformation resilience. However, each role has weaknesses and risks, making clear recommendations essential to mitigate potential harms.

Recommendations

Overall, AI systems must be used in reflective ways to maximise benefits while mitigating risks. Human oversight is essential to ensure that AI systems complement rather than replace human judgement. Transparency in AI-content generation and dissemination is needed to prevent misuse. AI’s persuasive capabilities should be monitored and designed to encourage reflection rather than passive acceptance. Education initiatives should promote AI literacy,

equip users with skills to assess AI outputs critically, and hinder the reinforcement of biases and stereotypes (more details in Table 1 and below).

Table 1. *The Seven Roles of AI in Misinformation Management - Strengths, Weaknesses, Opportunities, Threats and Recommendations.*

AI's Role	Strengths	Weaknesses	Opportunities	Threats	Recommendations
Informer	Rapid access, tailored explanations, multilingual search.	Bias, hallucinations, lack of source verification.	Improve speed and search.	Amplifies misinformation, manipulates information flows.	Regulatory oversight, transparency, digital literacy.
Guardian	Real-time monitoring, detects misinformation, “neutral” fact-checking.	Limited by training data, struggles with nuance, flags false positives.	Cross-language fact-checking, human-AI validation.	Potential misuse for political/economic influence.	Bias checks, trusted oversight, critical engagement.
Persuader	Persuasive at scale, counters conspiracy beliefs, enhances consensus.	Risk of manipulation of humans, depends on training data biases.	Personalised interventions, support e.g. public health messaging,	Flooding of low-quality content, micro-targeted persuasion.	Regulate micro-targeting, ensure source credibility
Integrator	Facilitates debate, synthesises diverse perspectives.	Susceptible to algorithmic bias, may reinforce misinformation.	Support of democratic debate, improve	Can be used to manipulate for antidemocratic purposes.	Regulatory oversight, human-AI collaboration.

			decision-making.		
Collaborator	Enhances efficiency and supports research.	Opaque reasoning creates illusion of understanding.	Enhances information access, processing and production.	Users overestimate AI reliability; overreliance damages critical thinking.	Encourage reflective use and AI literacy, ensure transparency.
Teacher	Provides scalable support, rapid individual feedback.	Bias, hallucinations, lack of contextual understanding.	Supports educators, improve access, refine comprehension.	Passive, uncritical learning, reinforcement of laziness and mainstream views.	Maintain human oversight, integrate AI literacy, restricted and reflective use.
Playmaker	Creates interactive learning experiences, adaptive storytelling.	Lacks creativity, reinforces stereotypes, requires human refinement.	Educational games can boost critical thinking, automates game development, enhances accessibility.	Biased content, ethical concerns and copyright risks.	Ensure fairness, human oversight in design.

The strengths, weaknesses, opportunities, and threats (SWOT) of AI roles

The following sections examine in greater detail AI's roles in relation to misinformation detection and dissemination.

The Informer

Strengths: AI enables efficient multilingual searches across vast multimodal datasets (3–8). AI systems can be accurate for factual queries—insofar as training data are accurate—and tailor explanations to different levels of complexity, enhancing accessibility. *Weaknesses:* Factuality is a challenge, as AI struggles with source evaluation and often generates information without proper verification (9–11). Hallucinations remain an issue, alongside AI's difficulties with recognising and correcting its own errors (12). Biases embedded in training data and fine-tuning processes can amplify stereotypes and skewed perspectives (13). Many commercial models lack transparency regarding their training data and reasoning. Susceptibility to intentional or unintentional prompt manipulation exacerbates misinformation risks (14, 15). *Opportunities:* AI integration with complementary tools could improve search accuracy (e.g., deploying localised AI models on personal devices to incorporate relevant context information). AI systems could help people find credible news (16). Human-in-the-loop systems can support expertise, ethical judgment, and nuanced decision-making in contexts where AI alone might fall short (17–20). *Threats:* AI lacks true expertise and may rely on dubious sources and false narratives, especially where data voids exist (21–26). AI can create illusory understandings while cementing misconceptions (27) and perpetuating biases (28, 29). Perceiving AI systems as authoritative or unbiased could increase susceptibility to flawed content (30–32). Control over AI systems confers significant power over information flows, raising concerns about manipulative or biased influence. Ethical and legal risks complicate AI deployment, with proprietary models potentially exposing copyrighted or personally identifiable content (7, 33–35). Low-quality AI content threatens search reliability

as models increasingly reference synthetic data (36). *Recommendations:* These threats highlight the need for regulatory oversight, transparency, and responsible use. Regulatory frameworks should mandate responsible AI development and deployment, discouraging models that promote unreliable information (37). Quality-control safeguards, including credibility-based ranking algorithms, human-in-the-loop systems, and regular bias checks, are required to mitigate misinformation amplification (38). Transparency should be prioritised, including disclosing training-data sources, labelling AI-generated content, and facilitating independent audits and public scrutiny. Legal protections against the misuse of personal data and copyrighted material by AI models could help ensure ethical AI deployment. Digital-literacy initiatives promoting effective AI use, with equitable access, are required.

The Guardian

Strengths: AI systems can rapidly analyse information, monitor platforms in real-time (39, 40), and produce convincing counter-messaging (41, 42), facilitating the identification and correction of emerging misinformation (30, 38, 43, 44). *Weaknesses:* AI's potential as a "Guardian" is limited by training data; AI has blind spots in certain areas and trouble detecting new misleading tactics. It also struggles to recognise misinformation that is subtle or contains logical fallacies (45–47). AI may erroneously flag accurate content as it lacks cultural, contextual, and historical awareness and understanding of literary devices (e.g., sarcasm) (48, 49). *Opportunities:* As AI systems are language-agnostic, they could be used to automatically detect and fact-check misinformation across countries and domains through browser plug-ins or online platforms (50, 51). This may be optimised through human-in-the-loop systems, with fact-checkers validating AI assessments. AI-generated corrections of political misinformation may be particularly useful, as they can be perceived as neutral, reducing motivated reasoning (52–54). *Threats:* Organisations could use fact-checking

systems to benefit their political or economic interests. Reliance on AI systems to detect and correct misinformation could evoke a false sense of security and reduce critical analysis.

Recommendations: Regulation could require platforms to conduct regular bias checks and apply system debiasing where necessary (38). Trusted organisations should invest in AI capabilities for combating misinformation and employ initiatives without being deterred by concocted debates about freedom of speech and truth arbiters, as misinformation can often be objectively and reliably identified (55). To avoid blind trust of AI systems, educational interventions that foster critical engagement must be developed.

The Persuader

Strengths: AI systems can rapidly synthesise diverse knowledge sources to generate persuasive messages at scale (26, 41, 56, 57), even in domains perceived as hallmarks of human interaction (e.g., social dilemma situations requiring expressions of empathy) (41). As a “Persuader”, AI can help counter conspiracy beliefs through tailored, evidence-based conversations (58). *Weaknesses:* People may show reactance to AI-generated persuasion in some contexts (41). AI biases and sensitivity to prompt engineering, combined with its limited ability to recognise its own limitations, can lead to highly persuasive yet misleading messages (21–25). *Opportunities:* The “Persuader” could amplify accurate information in important domains including public health and reinforce perceptions of consensus to serve the public interest. AI can also produce personalised content to enhance persuasion (57, 59–61). *Threats:* Large-scale content generation may be used to “flood the zone” with low-quality information. Persuasive messages, especially if micro-targeted, present a substantive threat if used by malicious actors against vulnerable populations (60, 61). Low AI literacy may also increase susceptibility to persuasive attacks due to uncritical receptivity (62). *Recommendations:* Given that persuasion depends on multiple factors including source credibility (63, 64),

message repetition (65, 66), perceived consensus (67, 68), communication channel (69, 70), and level of transmission (71), information sources must be made explicit. However, source transparency alone is likely insufficient (24, 26, 60) and must therefore be accompanied by appropriate regulation of micro-targeting in sensitive domains (e.g., elections).

The Integrator

Strengths: The “Integrator” may facilitate democratic deliberation by helping individuals find common ground in divisive political debates (72, 73). While humans show bounded rationality and biased reasoning (77–80), AI can rapidly integrate vast amounts of data across languages and modalities, identifying patterns and trends. *Weaknesses:* Data integration can be skewed by biases in training data, algorithm design, fine-tuning, and feedback loops (20–25). *Opportunities:* AI-generated summaries of debates can be perceived as superior and more balanced than human-mediator statements, which at scale may be able to support democratic deliberation (72). Better information synthesis may provide a foundation for improved policymaking. *Threats:* Integration algorithms could be manipulated to encourage pre-ordained outcomes by malign actors and facilitate antidemocratic discourse. Aggregated opinions may exert normative pressure on people to conform (78). *Recommendations:* Regulatory oversight may be required to ensure careful engineering and deployment of AI “Integrators”, using a human-in-the-loop approach.

The Collaborator

Strengths: AI systems can be useful as guides, coaches, and research assistants (79, 80), including in the context of information evaluation. *Weaknesses:* AI-systems often operate as black boxes, making it difficult to assess how they classify information or detect misinformation (80). The “Collaborator’s” reliability depends on the availability of sufficient

high-quality data and precise instructions. Quick and well-formulated AI responses can create a false sense of accuracy, leading users to accept misinformation without scrutiny (81–83) and overestimate their knowledge (27, 80). *Opportunities:* In academic and investigative settings, AI could assist by identifying patterns and expanding access to diverse sources of information (28, 84, 85). AI-driven tools could enhance metacognitive skills by supporting interactive learning processes and boosting reflection on information quality (80, 86). *Threats:* The effectiveness of AI “Collaborators” remains largely untested in the context of misinformation. Risks of misdirection exist particularly for users with limited knowledge and AI vigilance (80, 87, 88). Over-reliance on the “Collaborator” may reduce users’ ability to critically assess information, leading to skill erosion over time (89, 90). Furthermore, AI systems can have a tendency to feed users content that aligns with their pre-existing beliefs rather than challenging them (9). The cognitive demands of critically assessing AI-generated outputs may also lead to cognitive overload in users unfamiliar with the topic area and misinformation-detection strategies (80). *Recommendations:* AI-human collaboration must be approached with critical reflection. AI should act as a co-pilot rather than a truth arbiter replacing critical human judgment, encouraging users to evaluate its outputs rather than accept them uncritically. AI-literacy programs should emphasise AI vigilance and metacognitive support (80, 91). AI collaboration must be transparent, ensuring users understand its strengths and limitations (80, 92).

The Teacher

Strengths: AI may be able to facilitate education against misinformation, but the evidence available to date is scarce and mixed (93). *Weaknesses:* AI lacks contextual understanding, soft skills, and ethical reasoning, making it unreliable as a “Teacher” (e.g., misinterpretation of complex arguments or satire) (2, 96). *Opportunities:* AI offers scalability and rapid and

individual feedback (95, 96). Virtual agents could support media literacy (97–99), and AI-tools can simplify complex concepts and refine comprehension (100). With improved reasoning transparency, AI may enhance students’ ability to engage with information critically (101). *Threats:* AI in education risks fostering passive learning, viz. reliance on AI-generated answers rather than active task engagement (89, 102), which may lead to decline in cognitive skills. AI can reinforce mainstream perspectives over innovative ideas (103), subtly making students more susceptible to biased narratives. *Recommendations:* AI should serve as a teaching aid rather than a substitute for educators, maintaining a human-in-the-loop approach and a human-centred mindset in education (104, 105). Implementation in schools must be accompanied by AI-literacy programs that teach critical evaluation, bias recognition, and effective prompting (80, 94, 106). AI-driven manipulation, emotional profiling, and social scoring in schools should be prohibited (107).

The Playmaker

Strengths: Serious games have shown potential to foster fact-checking skills and critical thinking (108–111). *Weaknesses:* AI systems struggle with logical reasoning and consistency, leading to unpredictable content generation in live gameplay (112). Storytelling can be negatively affected by group stereotypes (29), AI’s tendency to lose track of long-term interactions due to limited memory (113), and lack of creativity and depth. *Opportunities:* AI can streamline game development by automating level design, narrative generation, and character creation, making serious-game development more feasible (115, 116, 117). Additionally, AI can make games more accessible through real-time translation and adaptive difficulty (117). AI-assisted simulations could also help train educators to counter conspiracy theories in classrooms (97). *Threats:* Over-reliance on the “Playmaker” risks producing generic, biased, and unoriginal content, reducing the effectiveness of serious games in

fostering critical thinking (118, 119). Ethical and legal concerns remain unresolved, including displacement of human designers, concerns related to AI-driven monetisation and potential manipulation of players, and copyright issues. *Recommendations:* AI-generated game content must prioritise accuracy, fairness, and inclusivity (116). Developers should ensure human oversight in AI-assisted game design to mitigate bias and maintain ethical standards. Players should be made aware of algorithmic, cultural, and economic biases embedded in AI-generated content.

Limitations

Research at the nexus of AI, misinformation, and human behaviour is still emerging. Studies often lack long-term validation and rigorous controls, rely on small samples, self-report measures, and short-term assessments. More rigorous studies, including randomised controlled trials, are needed to better understand AI's strengths, weaknesses, opportunities and threats.

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